



BPS5231

Artificial Intelligence

For

Sustainable

Building

Design

L6

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Assistant Professor

L06.1

Simulations

Rhino
ClimateStudio

L06.2

Simulation Game

L06.3

Surrogates

Surrogate/Meta Models

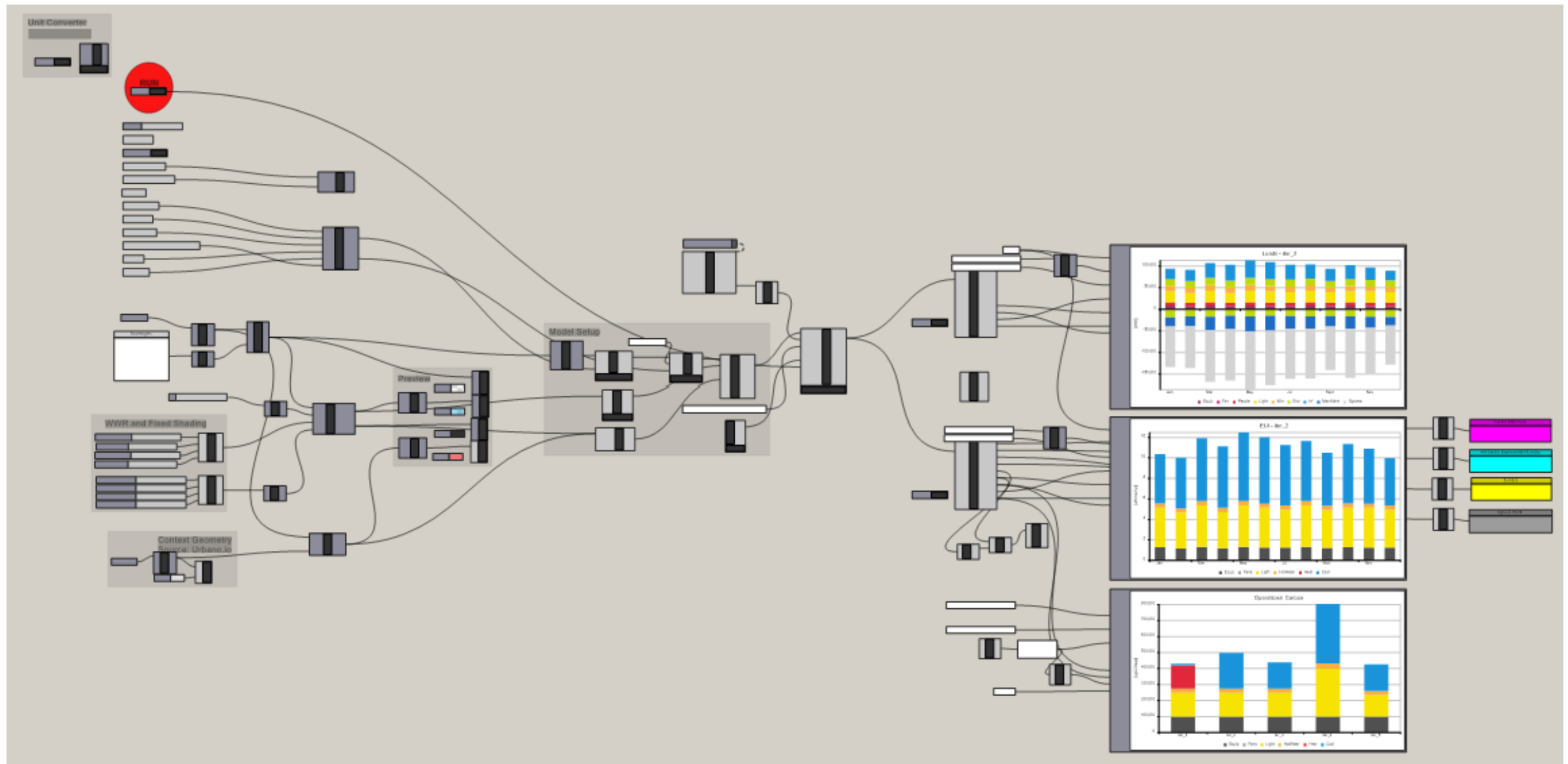
SIMULATION GAME

SIMULATIONS

GAME

SURROGATES

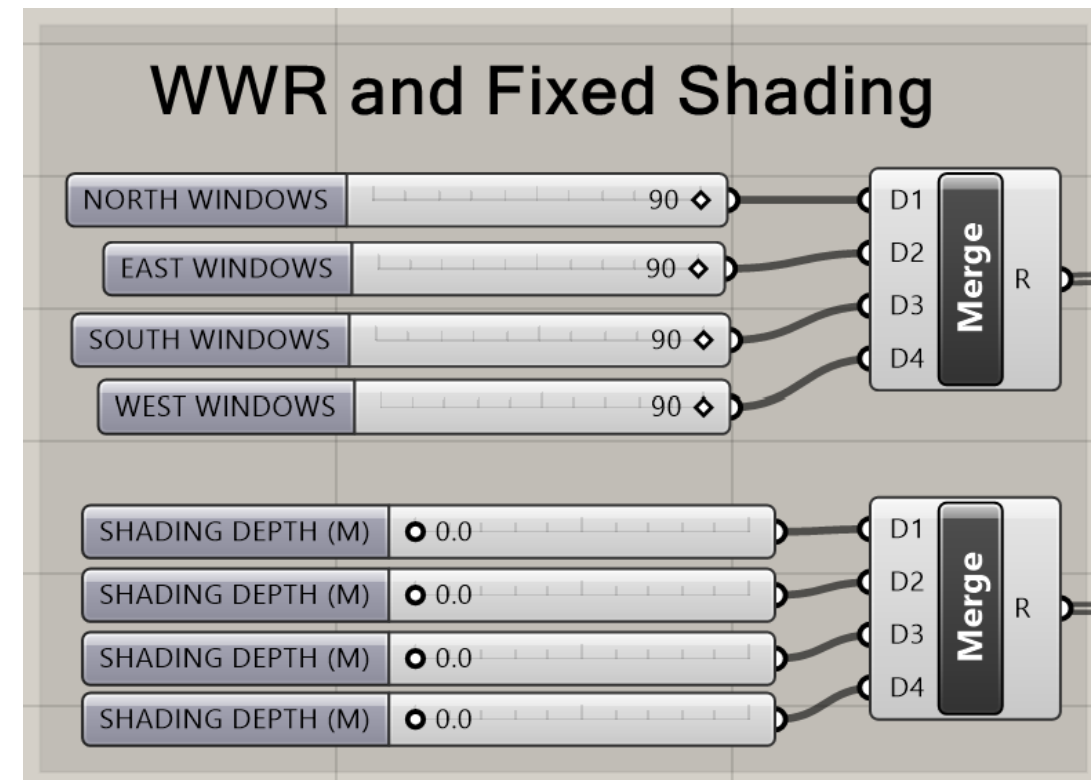
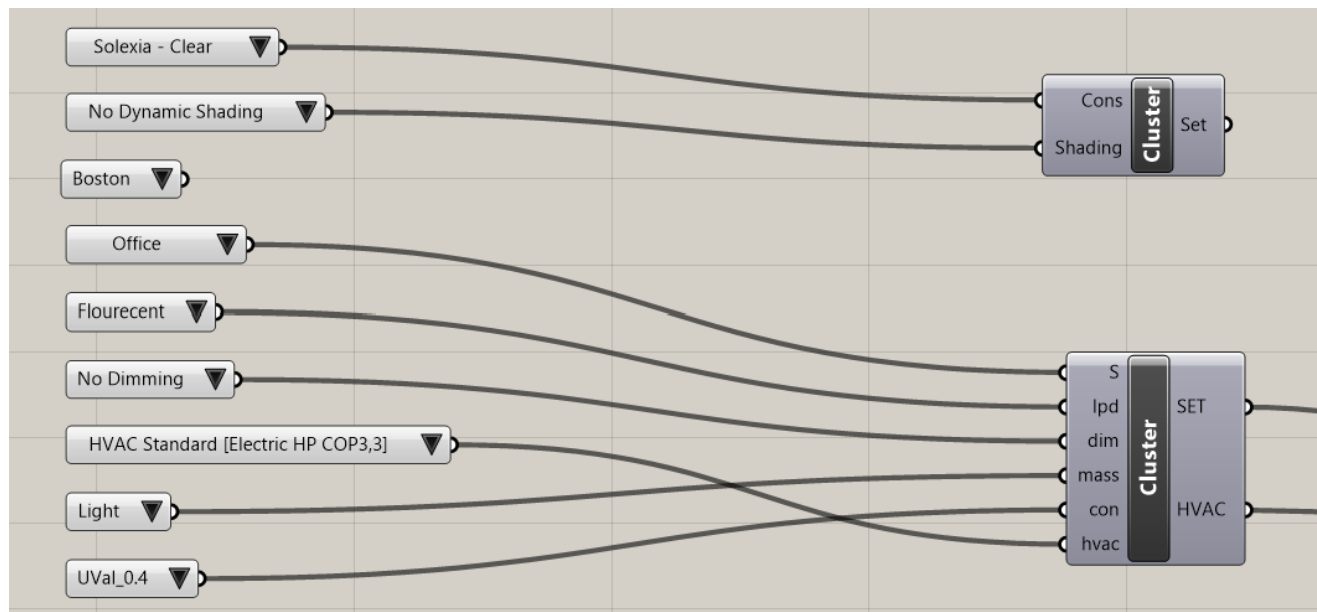
Simulation Game



SIMULATIONS

GAME

SURROGATES



Google Colab
<https://bit.ly/BPS5231-L6-SimulationGame>

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Surrogates

Surrogate/Meta Models

The Challenge

Modern engineering relies heavily on complex computer simulations (Energy+, CFD, FEA, thermal analysis etc)

Traditional approaches are computationally prohibitive for:

- Design optimization
- Uncertainty quantification
- Sensitivity analysis
- Real-time control systems

The Solution

- Replace expensive simulations with fast-running approximations
- Maintain accuracy while reducing computational cost by orders of magnitude
- Enable extensive design space exploration

In machine learning, a surrogate model is a computationally inexpensive, data-driven approximation of a complex, expensive, or time-consuming model (like a simulation) that it replaces.

These "meta-models" are trained using input-output pairs from the original model and then used to quickly predict outcomes for new inputs, speeding up design cycles, optimization, and analysis in various fields, including engineering, materials science, and environmental modeling.

What is surrogate modeling?

An engineering method used when direct measurement or computation of an outcome is difficult, expensive, or time-consuming

Creates fast-running mathematical approximations (surrogates) that mimic the behavior of complex simulations or experiments

Can be used for design optimization, uncertainty quantification, and sensitivity analysis

Also known as metamodeling or emulation

Why use them?

Speed and efficiency

They provide very fast predictions, often in seconds, which is crucial when dealing with simulations that can take hours or days

Cost reduction

By avoiding repeated expensive simulations, they lower computational costs

Rapid Design and Optimization

They enable faster iteration in design processes, allowing for quicker testing of new ideas and optimization.

Handling Complex Systems

Surrogate models excel at approximating complex or non-linear relationships where the fundamental model might be poorly understood or too complex to derive directly.

Sample Workflow

1. Sampling & Data Collection

- Design of Experiments (DoE) to select training points
- Latin Hypercube Sampling for space-filling designs
- Run high-fidelity simulations at selected design points

2. Model Construction

- Choose appropriate surrogate model type
- Train model using input-output pairs
- Optimize model parameters

3. Validation and Assessment

- Cross-validation techniques
- Error metrics (RMSE, MAE)
- Residual analysis

4. Active Learning (Maybe)

- Iteratively add new training points
- Focus on regions with high uncertainty or interest
- Improve model accuracy progressively

Building and Urban Energy Systems Examples

Building Performance

- Energy consumption prediction
- HVAC optimization
- Daylight and thermal comfort analysis

Urban Scale Modeling

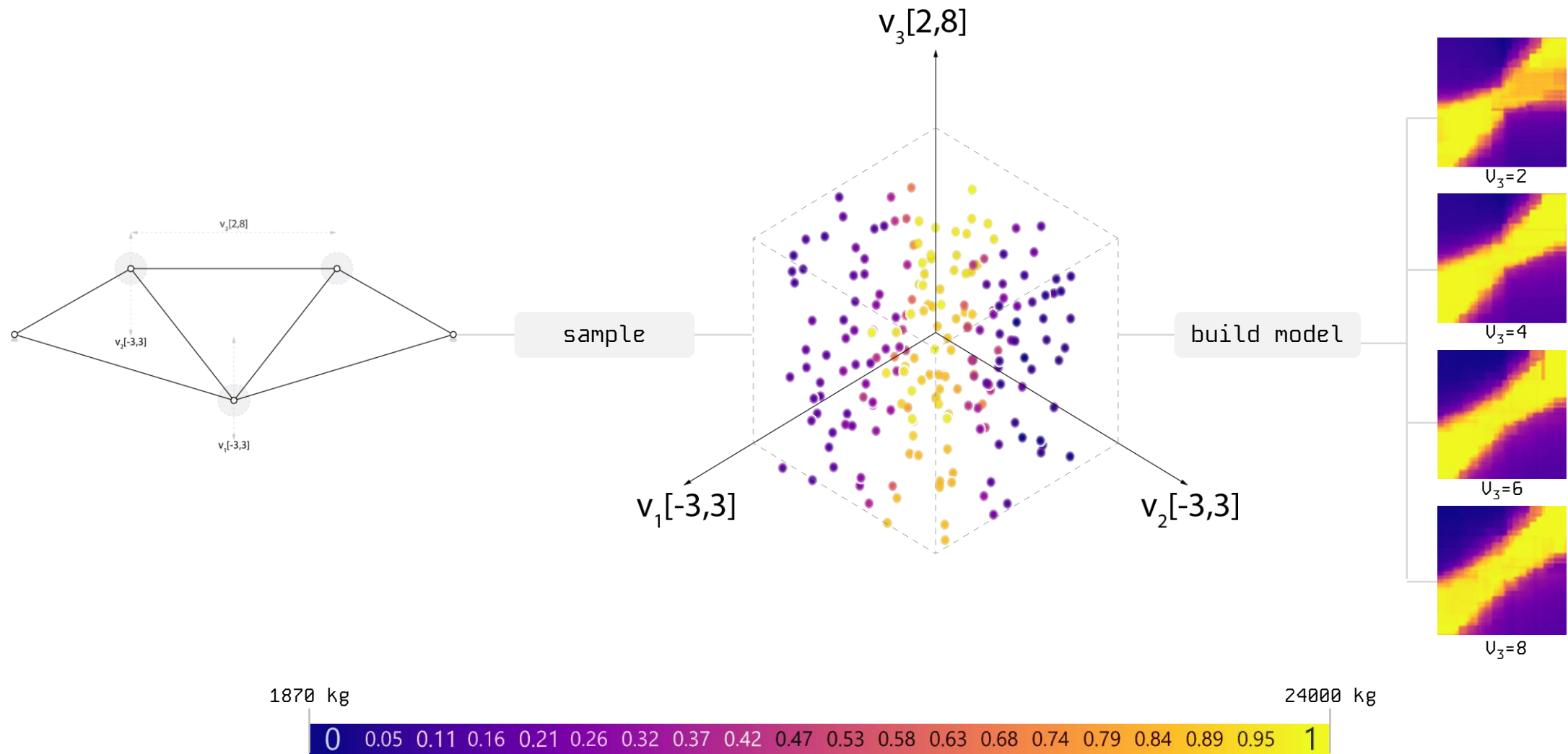
- Urban building energy use
- Urban heat island effects

Multi-physics Integration

- Coupled thermal-airflow-lighting models
- Building-grid interactions
- Occupant behavior modeling

Research Frontiers

- Physics-informed machine learning approaches
- Neural radiance fields for urban modeling

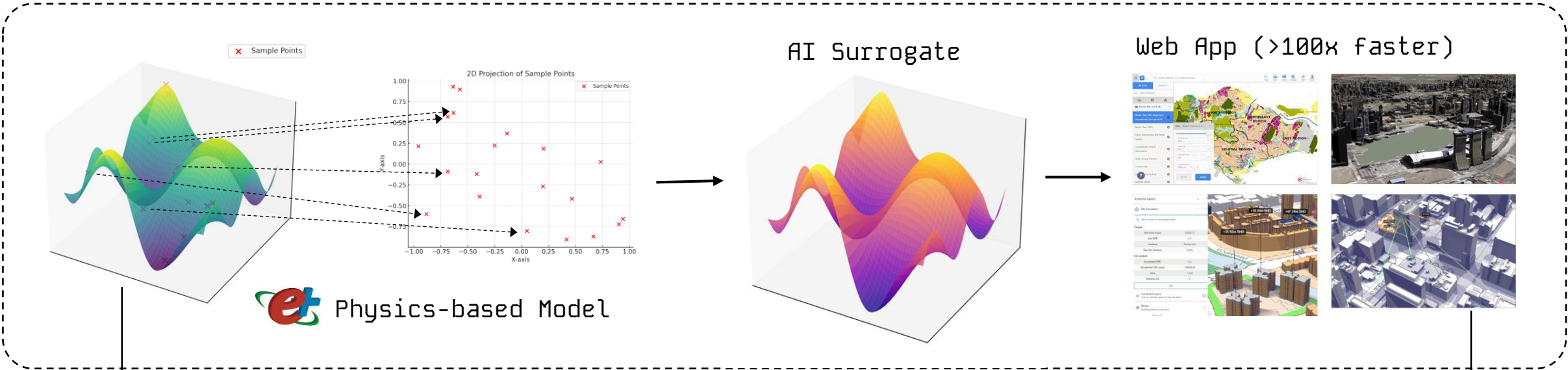


Source: Renaud Danhaive

SIMULATIONS

GAME

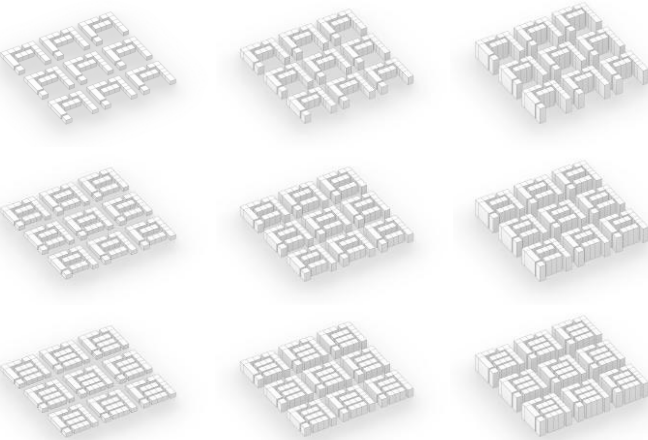
SURROGATES



Input Parameters	
General	
1	Building Program (Type)
2	Building Age (Years)
3	Building Gross Floor Area (m ²)
Structural, Enclosure, and Internal Partitions	
4	Floor-to-floor- height (m)
5	Perimeter zone depth (m)
6	Wall construction (W/m ² K)
7	Roof construction (W/m ² K)
8	External wall type (Type)
9	Internal walls/partitions (Type)
Fenestration	
10	Fenestration type (Type)
11	Fenestration shading (Type)
12	North window-to-wall ratio (%)
13	South window-to-wall ratio (%)
14	East window-to-wall ratio (%)
15	West window-to-wall ratio (%)
16	Window frame conductance (W/m ² K)
17	Glazing type (Type)
18	Window leakage (ACH)
19	Natural ventilation (Yes/No)

Heating, Ventilation and Air-Conditioning	
20	Thermostat set-point (°C)
21	Core outside air flowrate (ACH)
22	Perimeter outside air flowrate (ACH)
Density and Power	
23	Core occupant density (pp/m ²)
24	Perimeter occupant density (pp/m ²)
25	Core equipment power (W/m ²)
26	Perimeter equipment power (W/m ²)
27	Core lighting power (W/m ²)
28	Perimeter lighting power (W/m ²)
Schedules	
29	Occupancy schedule (h/wk)
30	Equipment usage (h/wk)
31	Lighting usage (h/wk)
32	Core outside air schedule (h/wk)
33	Perimeter outside air schedule (h/wk)

Energy Use Intensity
Carbon Emissions



L05.1

Supervised 5

Bagging

Random Forest

Boosting

XGBoost

L05.2

Visualizations

Exploratory Data Analysis

Communication

L05.3

Simulations

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